“Aprende ECG”: An educational device to improve medical students skills in ECG interpretation

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Abstract: The present work shows the development of an ECG simulator device. The proposed device can measure ECG in real time, simulate ECG signals from pathologies and display both of them in a display. It is portable, which makes it easy to carry, and integrates a GUI that makes it easy to use. The combination of all these features makes the device a solid option for students who want to improve their ECG skills interpretation.

Keywords: Electrocardiogram, ECG, education.

1 Introduction

The electrocardiogram (ECG) is one of the most widely used non-invasive diagnostic methods in clinical practice. A misreading of this signal can lead to morbidity and mortality, especially in cases of myocardial infarction, heart block and arrhythmias. Therefore, proper training in ECG interpretation is of paramount importance [1]. However, the training of medical students in ECG interpretation does not always meet expectations. ECG interpretation is a difficult skill to master and retain, indicating difficulties in the process of learning the technique. Many graduate students and professionals expressed discomfort in interpreting ECG waveforms and felt that the amount of time spent on this subject throughout their training was insufficient [2].

2 Problematic

The learning and interpretation of the electrocardiogram (ECG) present significant challenges globally and regionally. A study conducted at the University of Port Harcourt in Nigeria assessed 140 final-year medical students on their ability to place precordial leads and perform basic ECG interpretations. Alarmingly, 61.4% were unable to correctly place any leads, and only 7.8% managed to place all leads correctly. Regarding the interpretation of 20 ECG cases, 12 were not correctly interpreted by any student, and only 28 identified a normal ECG [3].

In Saudi Arabia, research involving 373 intern doctors revealed an average score of 5.4 out of 10 in an ECG interpretation questionnaire. Furthermore, 63.5% of participants considered their university training inadequate, reflecting insufficient preparation in this critical area [4].

A study in Malaysia with 324 medical students showed that over 50% had a negative attitude towards learning ECGs, citing difficulties in recalling concepts, understanding, and practical application [5].

In Latin America, a study at the University of Chile found that 81% of students deemed the teaching methodology inadequate for independently interpreting an ECG correctly. Similarly, in Peru, a study involving students from several universities revealed that only 66% correctly identified an acute myocardial infarction, with less than 45% recognizing other high-mortality pathologies [6][7].

Finally, in Arequipa, Peru, a study with 188 medical interns indicated that a concerning 37.77% had deficient ECG knowledge, underscoring the urgent need to enhance teaching and practice in this area to mitigate patient risks [8].

3 State of the art

WebECG, a novel ECG simulator based on MATLAB Web Figure, offers an educational tool accessible via web browsers. It approximates ECG signal components using triangular and sinusoidal waves, leveraging Fourier analysis. This simulator can generate signals from a healthy heart and simulate nine types of arrhythmias, modeling noise such as Baseline Wander, Powerline Interference, and muscle interference to mimic real signals [9].

In a separate study, a device was developed for simulating ECG signals to evaluate and calibrate cardiological monitoring equipment at Hospital Dos de Mayo. This addressed the lack of calibration certification, mitigating measurement risks. The device processes ECG signal databases from Physiobank, filters and segments the signals, and performs digital-to-analog conversions, conditioning analog signals for twelve leads. It operates within a frequency range of 30 bpm to 150 bpm and an amplitude range of 0.5 mV to 2 mV, adjusted via potentiometer. This device effectively simulates ECG waveforms, offering the unique advantage of adding signals from other parameters [10].

The Adafruit Menta ECG simulator, based on Arduino technology, replicates cardiac waveforms using three electrodes on the chest. It stands out for its low cost, approximately $60, and innovative methodology for generating ECG signals, which can be adapted for various waveforms. The waveform was digitized from an online image using Engauge software and processed with a custom Python program. This project emphasizes the educational benefits of DIY electronics, enabling learners and educators to build affordable ECG simulators [11].

Finally, the Cardiosim/Cardiosim I-plus/Promptcare simulator serves as a training tool for healthcare personnel, simulating human cardiac waves. It is available at a cost of approximately 300 soles, providing an accessible option for educational purposes [12].

4 Justification

The lack of resources and tools, as well as the complexity of analyzing and understanding the ECG, makes it a difficult task for medical students to understand. This can lead to bad diagnostics, misinterpretation of the ECG and incorrect treatments can be given to patients, increasing mortality and reducing life-quality.

5 Proposed solution

Based on these findings, we propose the design and implementation of an educational device aimed at enhancing medical students' proficiency in ECG interpretation.

This device features three operational modes, the three options appear on the touch and the users can select them by clicking on the corresponding buttons:

- Real-time ECG Measurement and Display: Using ECG electrodes, the device measures, conditions, and converts ECG signals. It then displays the measured ECG on-screen in real-time. It also calculates the beats per minute (BPM) and displays this value on the screen.
- Simulated ECG Signals for Pathologies: Pathological ECG signals obtained from the CONTEC MS400 Multiparameter Simulator are stored in the SD memory. In this mode, users can select a pathology and its simulation will be shown on the screen.
- Display of both Real-time ECG Measurement and Simulated ECG Signals for Pathologies: The users can select a pathology and the screen simultaneously displays both the measured signal and the simulated signal of a chosen pathology, allowing for comparison between the two.

6 Methodology

6.1 Idea validation interviews

We interviewed medical students and asked for their opinions on the proposed solution. They expressed a positive reception, noting that they liked the concept as it offered a portable and practical way to study ECGs, facilitating enhanced learning opportunities.

6.2 Electronic Design

For the electronic design of the proposed device, we took into account the desired features. In this case, the device should be capable of being able to make an ECG reading in real time to help students familiarize with electrode placement, and to allow students to visualize ECG signals that are related to cardiac disorders in order for them to compare on how much they differ with an ECG from a healthy heart. In that sense, the device should be able to store information about the pathological signals for future display. Furthermore, the device should always have this data at disposal for the user to retrieve it at any time.

Due to this proposed functionality, the device needs to effectively store information from a signal that simulates a cardiac disorder like arrhythmia, tachycardia, etc. Nonetheless, this function in particular will only be executed once for storing data. Having saved this data, the device should be able to display the signals for a pathology selected by the user. This process, in comparison to the previous one, will be the one that can be executed multiple times. The same idea applies for the real time ECG reading, as it is a process that is meant to be repeated by the user when using the device. Now, in regards to the signals,, the proposed system should be able to condition the input signal to suppress noise as much as possible, so that the resulting ECG signal (for both storage or display) is clear enough. Finally, as it is meant for students to familiarize themselves with ECG signals at any time, the device use should not be limited to being powered by the wall socket to initialize its systems, thus a battery powered approach is more appropriate. Therefore, the following flow chart is proposed:

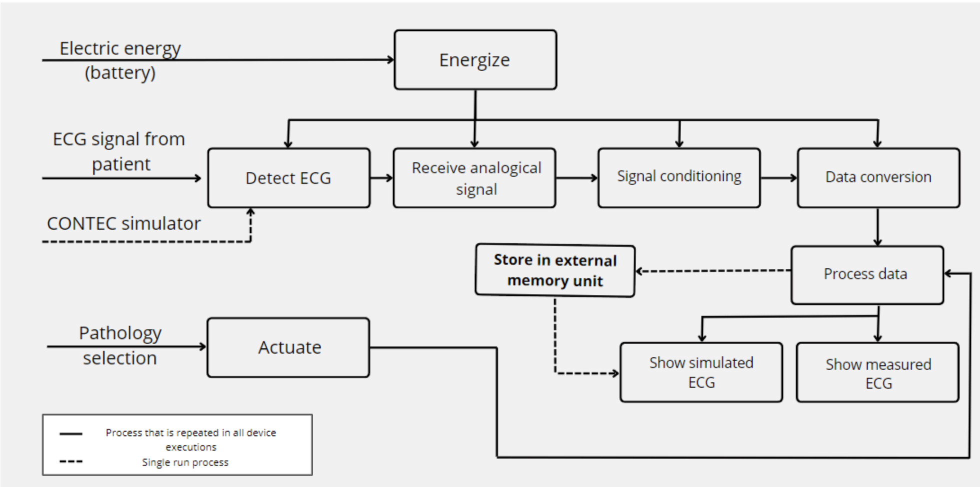

Fig. 1. General flow chart of the working protocol of the proposed system

It should be noted that, for practical purposes, the XSpace Bio 1.0 integrated circuit was used. This integrated circuit system already incorporated an ESP32 microcontroller, two AD8232 for ECG measuring, as well as multiple coding libraries that facilitate the task of ECG reading and processing. In addition, it has a USB type C port for programming the microcontroller as well as for energy input.

Having defined the working protocol, the electronic design itself could be addressed. In that sense, and according to the features the proposed device is intended to have, a PCB was designed via the EasyEDA platform. This environment allowed to translate the schematic design into a PCB.

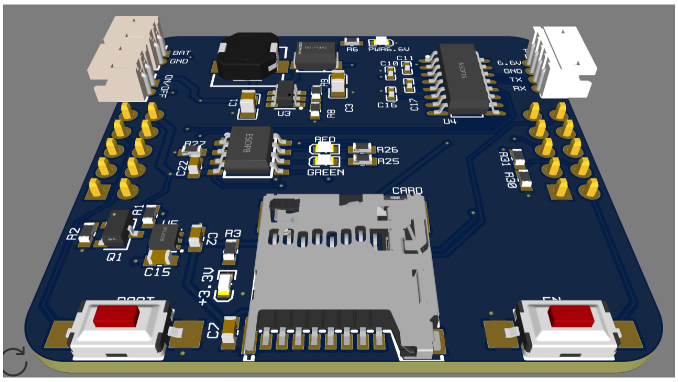

Fig. 2. PCB of the device

In the following figure, the PCB subsystems have been highlighted.

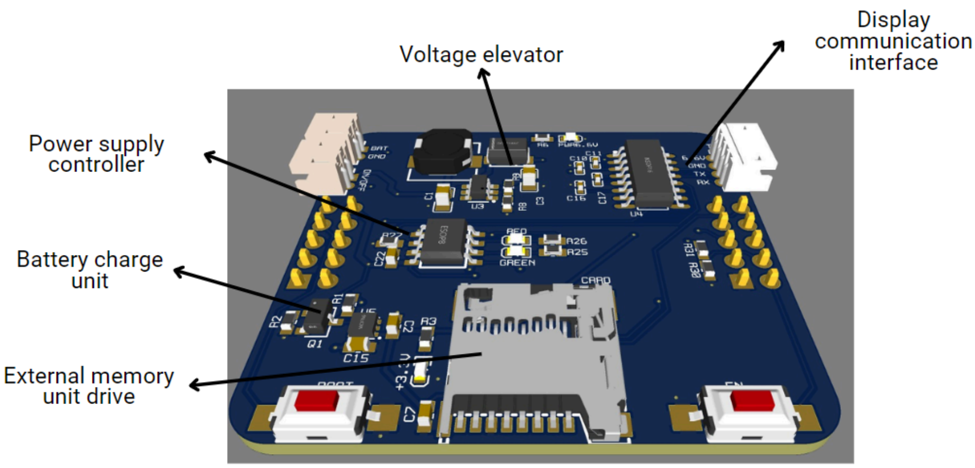

Fig. 3. PCB of the device and its subsystems.

We will briefly discuss the working principles of each subsystem of the PCB.

6.2.1 Battery charge unit

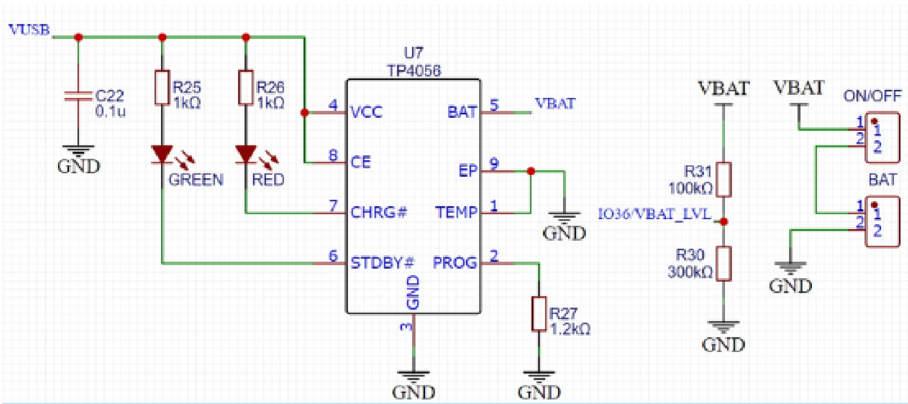

Fig. 4. Battery charge subsystem schematic

This subsystem allows batteries to be charged when the USB type C port of the XSpace is connected. When batteries are charging and are not at their top capacity yet, pin 7 goes into LOW state while pin 6 goes into high impedance, thus allowing current to flow only through the red LED path and turning it on. This gives the possibility to visually know when batteries are charging. When charging is done, pin 7 goes into high impedance while pin 6 enters LOW state. Now, current can flow through the green LED path, symbolizing charge completion.

6.2.2 Power supply controller

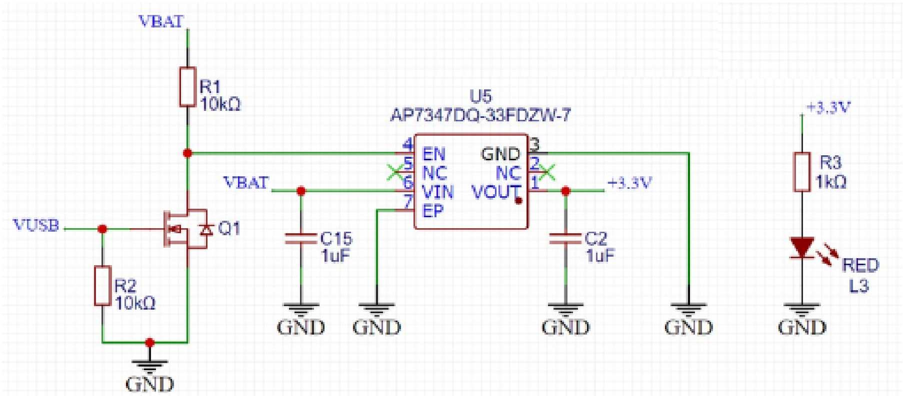

Fig. 5. Power supply subsystem schematic

The power supply subsystem allows the interchangeable feature for the circuit to be powered by the battery or by USB (when connected). Both the battery and USB port provide more voltage than required by the ESP32, then a voltage regulator to 3.3V is needed. We have labeled this regulator as U5. When connected via USB, mosfet Q1 activates and closes the circuit, disabling the regulator. This is important as the ESP32 regulates the 5V from USB internally. On the other hand, when not connected, Q1 opens the circuit, enabling U5 to regulate the battery voltage down to 3.3V.

6.2.3 External memory unit drive

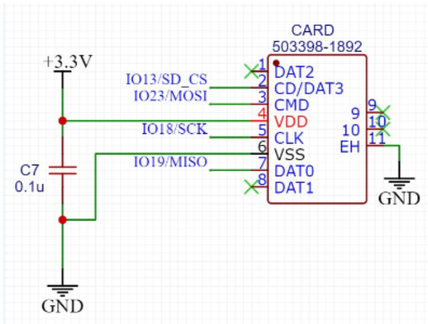

Fig. 6. External memory subsystem schematic

This subsystem allows the connection and communication between a micro SD card and the ESP32. Pin 2 serves as the chip select pin for a correct communication between master and slave device. On the other hand, pin 3 and 7 are connected to pin 19 and 23 of the ESP32 for MISO and MOSI functionalities respectively.

6.2.4 Display communication interface

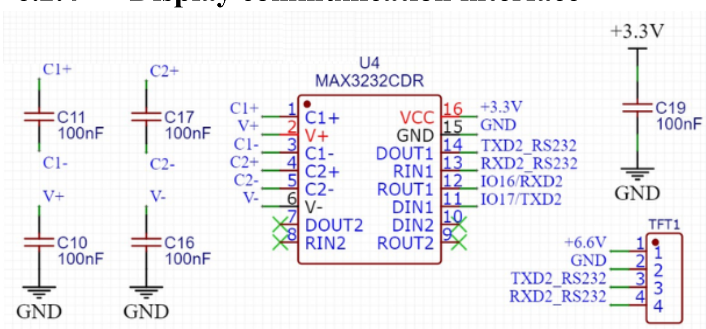

Fig. 7. Display communication subsystem schematic

Due to the RS232 communication protocol of the used display, it was needed to convert this protocol to one that could be interpreted by the ESP32 for proper communication between microcontroller and display. Therefore, the shown display communication interface subsystem was implemented to allow the connection between devices.

6.2.5 Voltage elevator

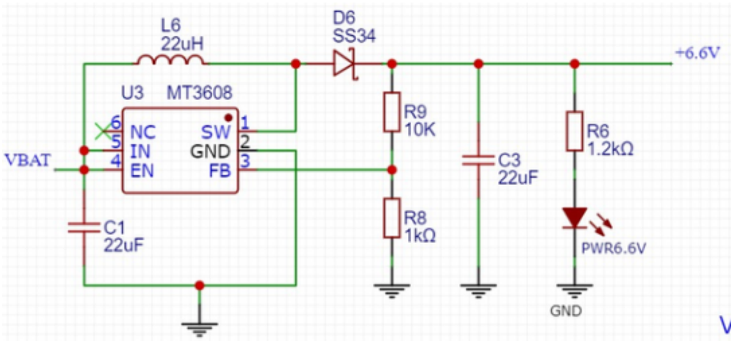

Fig. 8. Voltage elevator subsystem schematic

The used display required more voltage than the provided by batteries or USB, which is why a voltage elevator subsystem was added. The working principle is based on the charge process of both the inductor L6 and the capacitor C3. To prevent the output voltage from having a prominent ripple, a zener diode D6 was included in the design, thus keeping the voltage as stable as possible. We were able to control the output voltage of 6.6V by selecting R9 and R8 properly, as the product of this voltage divider allows the U3 integrated block to control the charge of the inductor in a way that 6.6V is generated at the output.

6.3 3D Model

The 3D model was designed in Inventor and consists of a box housing all electronic components. It includes a space designed to comfortably hold the screen at an ergonomic angle for the user. The model also features compartments for placing the 4 ECG electrodes used to obtain at the same time leads D1, D2, and D3. Additionally, it incorporates necessary openings for the on-off switch and a USB-C cable for battery charging.

The dimensions of the prototype are 16 x 25.5 x 12 cm, making it portable and easy to carry. Constructed from PLA, a lightweight material known for its portability, the prototype maintains a manageable weight suitable for practical use.

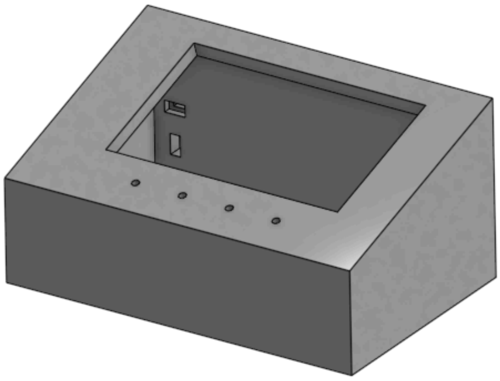

Fig. 9. 3D Model of the prototype.

7 Implementation

7.1 Hardware

7.1.1 PCB design

The PCB, as mentioned before, was designed using EasyEDA, with careful consideration of dimensions closely aligned with those of the XSpace Bio 1.0 for seamless connectivity.

7.2 Software

7.2.1 Firmware

We utilized the ESP32 microcontroller, which was included in the XSpace Bio 1.0, coding primarily in C. The main components of our firmware include reading ECG data from the AD8232 sensors, calculating beats per minute (BPM), reading data from an SD card, saving data on the SD card, and displaying the acquired data on the DWIN screen.

7.2.2 Screen configuration

To configure the DWIN 7 inch medical-grade screen (DMG80480K070_03WTR) we used DWIN's DGUS Software.

First, we designed the interface templates to be displayed on the screen. These templates included elements such as graphics, buttons, and areas to display data. Then we uploaded them to DGUS Software. In DGUS, we defined the interactive elements, such as buttons, and the coordinates where data would be displayed. We specified the exact coordinates where the relevant data, including beats per minute (BPM) and the electrocardiogram corresponding to leads D1, D2, and D3, would be shown.

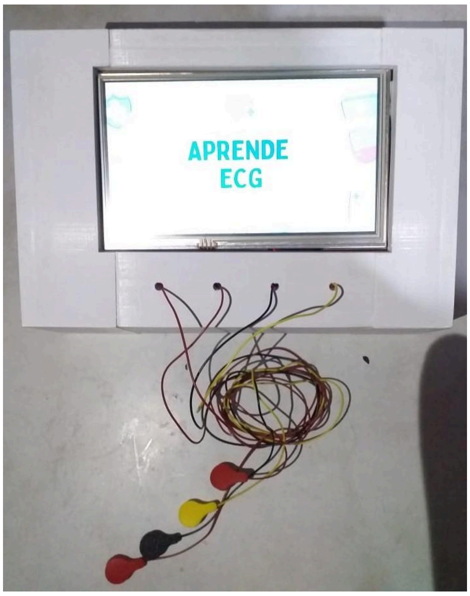

Fig 10. Prototype.

Fig 11. Prototype’s screen.

8 Results

8.1 Tests

For the prototype testing, we used the CONTEC MS400 Multiparameter Simulator to obtain the BPM (beats per minute) . Then, we compared the results obtained from the prototype with the ones given by the simulator.

Table 1. BPM Test

AVERAGE CONTEC BPM VALUE	AVERAGE PROTOTYPE BPM VALUE	AVERAGE ERROR
40	39	0.02
60	60	0.00
80	79	0.01
100	99	0.01
120	120	0.00
140	140	0.00
160	146	0.09

9 Conclusions

The average error in BPM calculation is very low (less than 3% up to 140 BPM), making the device highly accurate. Additionally, it effectively displays all three leads on the screen in real-time. The proposed device can be a powerful tool for medical students who want to improve their ECG interpretation skills and learning.

References

1. Alberna-Cardoso, A., Diéguez-Batista, R., Pérez-García, R.P.: Diagnóstico sobre la preparación de los estudiantes de Medicina para la identificación de alteraciones electrocardiográficas. In: Revista Médica Electrónica, vol.45, no.1. Enero-Febrero 2023 (2023). http://scielo.sld.cu/scielo.php?script=sci_arttext&pid=S1684-18242023000100124
2. Kashou, A.H., Noseworthy, P.A., Beckman, T.J., Anavekar N.S., Angstman, K.B., Cullen, M.W., Sandefur, B.J., Friedman, P.A., Shapiro, B.P., Wiley, B.W., Kates, A.M., Braisted, A., Huneycutt, D., Baranchuk, A., Beard, J.W, Kerwin, S., Young, B., Rowlandson, I.,

Knohl, S.J., O'Brien, K., May, A.M.: Exploring factors influencing ECG interpretation proficiency of medical professionals. In: Current Problems in Cardiology 48:101865 (2023). https://doi.org/10.1016/j.cpcardiol.2023.101865

3. Alikor, C. A., Onuwaje, P.: Competency in Electrocardiogram Interpretation and Placement of ECG Leads Among Final Year Medical Students of a Nigerian University. In: Cardiology Division, Department of Medicine, University of Port Harcourt Teaching Hospital, Nigeria, vol. 6, n.º 2, pp. 650-660 (2018). https://www.gazetteofmedicine.com/publications/COMPETENCY%20IN%20ELECTROCARDIOGRAM%20INTERPRETATION%20AND%20PLACEMENT%20OF%20ECG%20LEADS%20AMONG%20FINAL%20YEAR%20MEDICAL%20STUDENTS%20OF%20A%20NIGERIAN%20UNIVERSITY.pdf
4. Mousa, A.M.A., Alhubail, F.M., Almulhim, M., AlBeladi, B.A., Almulhim, N.A., Almulhim, A.A., Algouf, I.A., Khan, A.S.: Electrocardiogram Interpretation Competency of Medical Interns in Saudi Arabia: a Cross-Sectional Study. In: Cureus (2023). https://doi.org/10.7759/cureus.37557
5. Ohn, M., Souza, U., Ohn, K.: A qualitative study on negative attitude toward electrocardiogram learning among undergraduate medical students. In: Tzu-chi Medical Journal/Cí-jì Yīxué 32:392 (2020). https://doi.org/10.4103/tcmj.tcmj_91_19
6. Gómez, P.,, Valdés, C,, Yévenes, K., Castro, A.: Autoconfianza de los estudiantes de Medicina en su interpretación correcta de un electrocardiograma. In: Revista Hospital Clínico Universidad de Chile, vol. 27, no. 2, pp. 93–101, Jun. 2016 (2016). https://doi.org/10.5354/2735-7996.2016.70956.
7. Rubio, F., Haim, S.: Destrezas y habilidades en alumnos del último año de medicina para interpretar electrocardiogramas de emergencias cardiológicas. In: Upch.edu.pe (2017).
8. Cornejo Arias, J.S.: Nivel de conocimiento sobre la valoración de parámetros electrocardiográficos en estudiantes de medicina de último año, Arequipa 2020. In: Universidad Católica de Santa María (2020). https://repositorio.ucsm.edu.pe/items/305e621e-c547-447a-906e-05fa4f510b03
9. Güney, E., EkşI, Z., Çakıroğlu, M.: WebECG: A novel ECG simulator based on MATLAB Web Figure. In: Advances in engineering software, vol. 45, no. 1, pp. 167–174, Mar. 2012 (2012). doi: https://doi.org/10.1016/j.advengsoft.2011.09.005.
10. Cajahuaringa Salazar, J.E.: Diseño de dispositivo simulador de señales ECG aplicados a evaluación de equipos de monitoreo y registro de parámetros cardiológicos del Hospital Dos de Mayo. In: Utp.edu.pe (2020). https://hdl.handle.net/20.500.12867/3304
11. lynchzilla. Arduino Menta ECG Simulator project. In:GitHub (2024). https://github.com/lynchzilla/ecg_simulator/blob/master/Adafruit%20Menta%20-%20ECG%20Simulator.pdf
12. Ecg Simulator Cardiosim I /Cardiosim I-Plus / Promptcare, https://www.indiamart.com/proddetail/ecg-simulator-cardiosim-i-cardiosim-i-plus-promptcare-2849318349712.html, last accessed 2024/06/30.